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Integrated waveguide circuits for optical quantum computing

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Abstract: Although practical realisation of a fully functioning quantum computer is still a long way off, recent progress both experimentally and theoretically is paving the way for possible implementations. One of the leading approaches is quantum optics, where photons are used as carriers of quantum information, and are manipulated in both linear and non-linear optical circuits. More recently, advances in integrated quantum photonics are enabling the realisation of compact and stable quantum gates and circuits. This approach provides routes to enhancing the complexity of quantum optic experiments, and ultimately towards the development of advanced quantum technologies. This study presents an overview of recent developments in the field of integrated waveguide quantum circuits. Key building blocks required for the realisation of quantum circuits are presented, and a rudimentary version of Shor's quantum factoring algorithm is demonstrated. Planar waveguide quantum circuits provide a high-performance platform from which quantum technologies and experimental quantum physics using single photons can be developed, and a new generation of quantum information and computing devices can be monolithically integrated onto a single optical chip.

1 Introduction

Information is a key global resource, and continued growth of our global economy relies on an ever-increasing capacity to process and communicate information. Many multinational companies and governmental organisations now rely heavily on the ability to process large quantities of information, and the rapid rise of information-intensive Internet-based companies, such as Google Inc., demonstrates the clear need for increased information processing capabilities. Over the past half century, integrated electronic circuits based on classical physics have driven this information revolution. Through gradual miniaturisation, this technology has brought continued increases in capacity. This miniaturisation has reached so far that it is now possible to count the number of atoms across the smallest feature of a semiconductor transistor. At this scale, quantum mechanical effects are beginning to perturb the operation of the transistor, and thus the demand for higher performance is approaching the limits of classical physics. There are many different approaches currently under investigation to further push the limits of classical computing [1], but ultimately quantum mechanical effects will swamp the deterministic and predictable requirements of classical processing. An alternative approach is to use these quantum mechanical effects to our advantage, for instance, in the realisation of a quantum computer.

The power of the quantum computer hinges on the principle that information is physical [2], and as such it is possible to use the unique properties of quantum mechanics to encode, process and transmit information in ways that are not possible using classical systems. Quantum information science has shown that harnessing quantum mechanical effects can dramatically improve performance for certain tasks in communication, computation and measurement, and offers significant opportunities to revolutionise information and communication technologies [3].

2 Quantum computing

Today's computers process information according to the original hypothesis of Church and Turing [4]. These machines represent information in binary format, stored as 0s and 1s, and the computation operations that can be performed on these binary bits are limited by the laws of classical physics. At the heart of quantum computation, lies the ability to represent information as quantum bits (qubits), superposition states of both 0 and 1. Before measurement, the qubit can be considered to be in two states at the same time. Upon measurement the superposition state collapses into one of the base states (being 0 or 1 in this case) with a probability defined by the superposition state, and all quantum information is lost. For quantum systems with many qubits, the size of the superposition state increases with the number of qubits, such that a system with n qubits can be considered as being in 2^n states simultaneously. This is the key to the exponential computational power of a quantum computer, but this parallelism of processing can only be taken advantage of for a certain set of computation algorithms (and currently only a small set of algorithms have been identified). As such, it is unlikely that the quantum computer will be a replacement for classical computers but rather an addon, although a sufficiently powerful quantum computer would in theory be able to perform all of the tasks of a classical computer.

The first significant breakthrough demonstrating the computational advantage of a quantum computer over its classical counterparts came about in 1994, when the mathematician Peter Shor discovered that by applying the laws of quantum mechanics to computational problems, it was possible to factorise large integers with greater efficiency than with a classical computer [5]. Since then, there has been significant interest and intense effort aimed at the realisation of such a quantum computer. Although to date, only a handful of quantum algorithms have been discovered, they include important mathematical problems such as factoring, the quantum Fourier transform and search algorithms [5–7]. It is anticipated that as quantum technologies progress and quantum computing systems become available to a wider audience, the range of new applications will significantly increase [8]. One clear application for the quantum computer will be in the efficient simulation of large quantum systems, having wide implications in material development and nanotechnology.

The requirements to realise a quantum computer were put forward by Divenchenzo in 1998 [9]. Scalable physical qubits are required that can be well isolated from the environment (long coherence time), but also initialised, measurable and controllably interfaced to implement a universal set of quantum logic gates. In an attempt to fulfil this criterion, several schemes for quantum computing based on different quantum systems have been proposed, including electrons in atoms or ions, photons, magnetic flux, nuclear spins or magnetic resonance, solid state and superconducting systems [10]. Of all the systems proposed, photon-based quantum technology is a very promising candidate [11].

3 Photons for quantum computation

Photons are ideal carriers of quantum information: they have an extremely long coherence time and can be readily transmitted through free-space or fibre, making them the natural choice for quantum key distribution (QKD) systems, or the transmission of information between quantum processors. The quantum information can be encoded in any one of the numerous degrees of freedom that the photon has to offer. For example, linear polarisation is commonly used in bulk optic implementations, with the horizontal polarisation representing the $|0\rangle$ state and the vertical polarisation representing the $|1\rangle$ state (although any mutually orthogonal axes could be used). It is also possible to pack much more information into a single photon than just one qubit, by encoding information into several different degrees of freedom [12]. These include time-bin, path, frequency and orbital angular momentum, and, unlike polarisation, these additional degrees of freedom have an unlimited number of orthogonal states, allowing the possibility of encoding information in a superposition of more than two states.

A universal quantum computer could be implemented by using only arbitrary one-qubit operations and any non-trivial two-qubit gates [13], in much the same way that classical computers can be constructed from a very limited gate set. A two-qubit quantum logic gate requires qubits to interact with one another, and for photons this means that one photon should be able to induce a π phase shift on another photon. Unfortunately, photons do not readily interact with each other without some form of matter-mediated process, and even then the efficiency of photon–photon coupling through a non-linearity rarely exceeds 10^{-10} [14, 15]. To date, no known material has achieved a strong enough non-linearity to implement a phase shift allowing reversible interaction of two photons with high efficiency, despite significant effort in single atoms in high-finesse optical cavities, quantum-dots in cavities and electromagnetically induced transparency [15–18].

Considering the above, it would seem logical to assume that photons will be good for the communication of quantum information (i.e. linking quantum processors together), but would not be useful in quantum computation. However, in 2001, Knill *et al.* [19] proved that scalable all-optical universal quantum computation with simple linear optical elements, single-photon sources and single-photon detectors was feasible – thus negating the necessity to have a strong optical non-linear element. Known as linear quantum optics, the non-linearity required to make two photons interact is replaced by a measurement-induced non-linearity, through an irreversible change in the quantum state.

Realisation of a scalable quantum computer will require highly efficient single-photon sources, highly efficient single-photon detectors and low-loss linear optical networks that can control, manipulate and interfere single photons. The remainder of this paper focuses on the development of a photonic waveguide architecture that allows unprecedented control of photons in the quantum regime, while the development of single-photon sources and detectors is discussed towards the end of the paper.

4 Integrated waveguide quantum photonics

In addition to efficient single-photon sources and detectors, photonic quantum technologies will require sophisticated optical circuits involving high visibility classical and quantum interference. Prior to the work of Politi *et al.* [20, 21] in 2008 on integrated quantum photonics, quantum optic information science experiments were routinely performed using large-scale (bulk) optical components such as beamsplitters and mirrors, bolted to large optical tables with the photons propagating through air. These bulk circuits are inherently non-scalable and not sufficiently stable for reliable operation on a large scale. These limitations can be overcome by implementing all the basic functionalities of a bulk linear optic quantum circuit using planar lightwave circuit (PLC) technology. Such quantum optic integrated circuits not only reduce circuit size, but also allow stability and control of the optical path length previously unfeasible in large-scale bulk quantum circuits.

4.1 Qubit encoding

In free-space quantum optic circuits, polarisation is generally used to encode qubits, but in integrated waveguide circuits it is possible to use path encoding to represent quantum information, alleviating many of the problems associated with the control of polarisation. The qubit is encoded by an uncoupled pair of single-mode waveguides – know as the ‘dual rail’ representation of a qubit [22]. Direct relationship between quantum mechanics and the coupled wave equation used to describe the behaviour of PLCs has been established by Cincotti *et al.* [23], and in the dual-rail representation of the qubit the presence of a photon in the upper waveguide indicates the $|0\rangle$ state, and a photon in the lower waveguide indicates the $|1\rangle$ state. An arbitrary single photon qubit state is given as the superposition of these states: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of finding a photon in the upper and lower waveguide, respectively, and where $|\alpha|^2 + |\beta|^2 = 1$. Accurate control and manipulation of the qubits is necessary to realise useful quantum circuits, which can be achieved using combinations of beamsplitter and phase shifter. Fig. 1 shows the simple circuits required to perform the Hadamard operation and a rotation about the z-axis. Indeed, any elementary single-qubit unitary transform can be described using only waveguide couplers and phase shifters [24]. Optical networks for linear optical quantum information processing and, more generally, for a broad range of quantum optics experiments, can be built using just straight waveguides, bends, couplers and phase controllers.

4.2 Technology

PLC technologies developed by the telecommunications industry provide the perfect platform for the development of waveguide quantum optic circuits. Complex devices fabricated in material systems such as InP, GaAs, silicon and silica have been used to develop commercial components for transmitting, routing and detecting high-speed optical signals for data communication systems. PLCs have also been employed in single-photon QKD systems for use as time-bin entanglement circuits [25, 26].

The silica-on-silicon material systems is an ideal candidate for waveguide quantum circuits since it exhibits low waveguide losses ($<$0.1 dB/cm), couples efficiently to single mode optical fibre and is transparent over a broad range of wavelengths (from visible into the far infrared). Light is confined within the silica integrated photonic circuit by a core waveguide layer that has a higher refractive index than the surrounding cladding layer. A schematic of the silica-on-silicon waveguide structure is presented in Fig. 2, along with the mode field simulation.

Single-mode operation is essential to ensure perfect mode overlap between the different fields in the waveguides, so as to obtain good classical and non-classical interference. Using finite-element modelling techniques, the waveguide cross-section was designed to support a single propagating mode at a wavelength of 804 nm. With a core-cladding refractive index contrast of $\Delta n = 0.5\%$, the waveguide dimensions were calculated to be 3.5 μm × 3.5 μm. Optical components such as directional couplers and Mach–Zehnder interferometers (MZIs) were designed using beam-propagation modelling software. The devices were fabricated on a 4-inch silica-on-silicon wafer using standard optical lithography and dry etching techniques. Thermally grown undoped silica was used as the bottom cladding

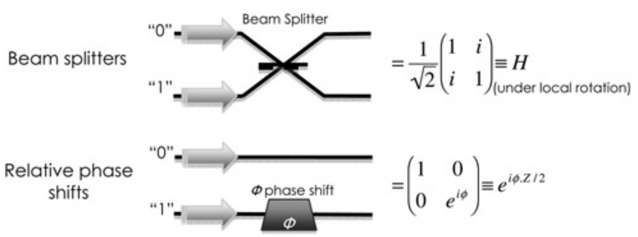

Fig. 1 *Dual-rail representation of a qubit*

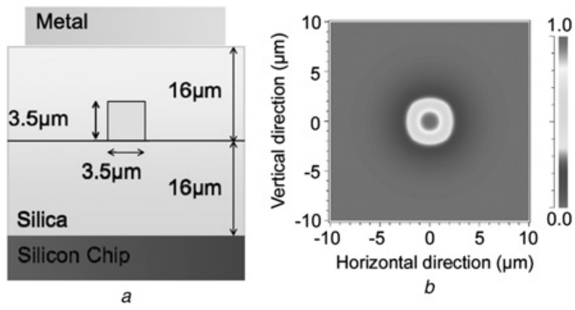

Fig. 2 *Details of the optical waveguide*

a Silica-on-silicon waveguide geometry
b Corresponding mode-field simulation

layer, and the waveguide core was formed from 3.5 μm thick layer of silica doped with germanium and boron oxides, deposited by flame hydrolysis. The top cladding layer, of thickness 16 μm, was overgrown with phosphorus and boron-doped silica. The refractive index of this layer was matched to the refractive index of the bottom cladding layer. Metal patterned on the top of the waveguide devices provided resistive heaters, allowing local control of the chip temperate, and providing a means to locally alter the refractive index.

A single wafer, containing hundreds of individual devices, was diced into several chips and the facets were polished to improve optical coupling from the optical fibre to the chip. Light was coupled in and out of the chips using single-mode fibre arrays, with an array spacing of 250 μm. A bright-light continuous-wave laser source was used to characterise the classical performance of the waveguide devices. A typical fibre-to-fibre coupling efficiency of 70% was routinely observed, and there was a 3% discrepancy between the designed and measured directional coupler coupling coefficients.

5 Quantum interference in waveguides

Quantum interference lies at the heart of linear quantum optics, setting it apart from the behaviour of classical optics. It describes the interaction of indistinguishable photons at a beamsplitter, and the first demonstration of this non-classical interference was presented by Hong *et al.* [27] in 1987. For integrated quantum photonics to be a viable alternative to bulk optics, it is essential that high visibility non-classical interference can be routinely achieved in waveguide circuits. This was first demonstrated by Politi *et al.* [20].

From a quantum information perspective, the operation of a simple beamsplitter can be described by the unitary mode transformation given by the matrix

$$\boldsymbol{U} = \begin{pmatrix} \sqrt{\eta} & i\sqrt{1-\eta} \\ i\sqrt{1-\eta} & \sqrt{\eta} \end{pmatrix}$$

where the beamsplitter reflectivity is given by η. Note that this matrix is equal to a single qubit rotation acting on a dual-rail path encoded qubit, represented in the computational basis. In waveguide photonic circuits, an equivalent to the beamsplitter, known as the directional coupler, can be realised by designing a device where the waveguides are in very close proximity to one another (Fig. 3*a*). In this situation, the evanescent fields from the

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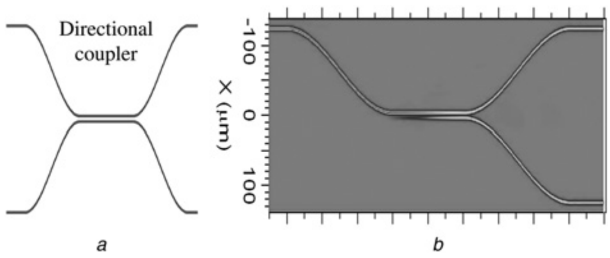

Fig. 3 *Integrated waveguide implementation of a directional coupler*

a Schematic diagram
b Optical field propagation simulation

waveguides overlap, and the light is able to couple from one guide to the next, as seen in the beam-propagation simulation in Fig. 3*b*. An arbitrary split ratio can be realised by varying the separation of the waveguides or the length of the coupling region.

When two indistinguishable photons simultaneously arrive at a beamsplitter whose reflectivity is 0.5, photon bunching occurs at the output ports due to quantum interference. The probability amplitudes of the indistinguishable photons destructively interfere for the conditions of transmission and reflection of individual photons, and the input state $|11\rangle$ is transformed to the output state $1/\sqrt{2}(|20\rangle - |02\rangle)$. Upon measurement, there are either two photons in the upper path or two photons in the lower path. The quality of the quantum interference is quantified by the visibility $V = (C_{\max} - C_{\min})/C_{\max}$, where $C_{\max}$ is the classical rate of photon detection in the absence of quantum interference, and $C_{\min}$ is the experimentally measured rate for the conditions of maximum quantum interference.

To demonstrate the suitability of integrated directional couplers as devices for quantum information science experiments, a simple two-photon experiment was performed (Fig. 4). Photon pairs were generated using parametric down conversion in a bismuth triborate (BiBO) non-linear crystal, and were collected using polarisation maintaining fibre at opposite points along the cone of light generated from the crystal. The photons were then launched into the directional coupler using single-mode fibre arrays. The coincident counts at the output ports of the direction coupler with respect to the arrival time of the two photons were measured. At a near-zero delay in relative photon arrival time, the coincidence detection rate dropped by $94.8 \pm 0.5\%$, demonstrating the high levels of quantum interference of photons in this integrated waveguide architecture.

Quantum interference has also been investigated in alternative waveguide technologies, with results demonstrated in direct write waveguide technologies in silica [28, 29]. This approach has potential advantages in terms of rapid prototyping and the possibility to realise three-dimensional structures.

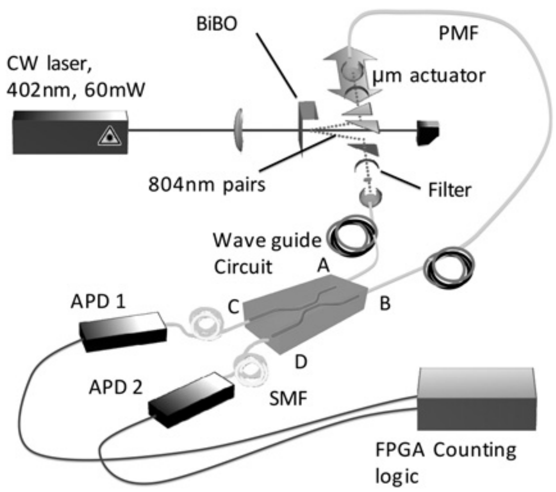

Fig. 4 *Schematic of experimental setup to characterise a directional coupler*

6 Control and manipulation of multi-photon entanglement

Precise control and preparation of qubits are of fundamental interest to the development of quantum technologies. Manipulation of a path-encoded qubit requires control of the relative phase ϕ between the two optical paths, and also the amplitude in each path. Such an operation can be achieved using the circuit shown schematically in Fig. 5, which comprises two directional couplers (reflectivity $\eta = 0.5$) and a phase shifter in a simple MZI configuration. The device performs the unitary operation $U_{\rm MZI} = H\, e^{i\phi\sigma_Z/2} H$, where H is equivalent to a Hadamard operation. Each optical mode at the inputs is transformed into a superposition state across modes c and d by the directional coupler, for example, $|1\rangle_a|0\rangle_b \rightarrow 1/\sqrt{2}(|1\rangle_c|0\rangle_d + |0\rangle_c|1\rangle_d)$. The phase shifter in the lower arms controls the relative phase between superposition states before the two modes are recombined at the second coupler. In conjunction with an additional phase shifter in the output arm, this circuit would apply the operation $U = e^{i\theta\sigma z/2} H\, e^{i\phi\sigma z/2} H$ (where θ and ϕ are two real free parameters, H is the Hadamard operation and sigma z is the third Pauli operator), thus allowing the possibility to map arbitrarily from the logical qubit $|0\rangle$ to the generalised state $\cos(\phi/2)|0\rangle + e^{-i\theta}\sin(\phi/2)|1\rangle$ and also from the computation basis $\{|0\rangle, |1\rangle\}$ to any other given basis.

Different physical effects and materials can be used to obtain the control of optical phases inside an integrated circuit. However, in the case of silica, the simplest method to control the propagation is via the thermo-optical effect, and this approach has been used for quantum waveguide experiments in both silica-on-silicon [30] and directly written waveguide devices [29]. By using resistive elements fabricated onto the surface of the device (see Fig. 2*a*), the temperature T of the waveguide structure directly beneath the resistor can be locally raised. The change in the temperature of the waveguide provides a change in the refractive index n of the order $dn/dT = 10^{-5}$K [31], giving rise to a change in the optical phase.

By varying the phase, the interferometer circuit represented in Fig. 5 can be used as a variable beamsplitter, allowing one to dial up any desirable reflectivity required. When acting on

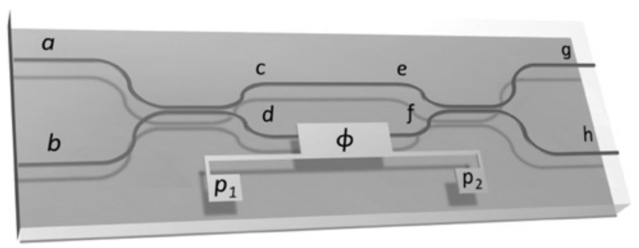

Fig. 5 *MZI with variable phase shifter*

the arbitrary superposition of the two input paths, the unitary operation of this device is given by the transfer matrix [32]

$$U = \begin{pmatrix} \sin(\phi/2) & \cos(\phi/2) \\ \cos(\phi/2) & -\sin(\phi/2) \end{pmatrix}$$

Compared with a standard beamsplitter, this corresponds to reflection and transmission coefficients of $R = \sin^2(\phi/2)$ and $T = \cos^2(\phi/2)$, respectively. Quantum interference measurements were performed using this device for a range of different effective reflectivity's by varying the phase within one arms of the interferometer [30]. In each case, two photons were launched into the inputs *a* and *b*, and the relative arrival time of the photons was scanned while monitoring the rate of coincidence detections at the output ports. The visibility of the non-classical interference dip for each reflectivity is plotted in Fig. 6*a*. The insets show two examples of the non-classical interference plots, with Fig. 6*b* showing a visibility of $98.2 \pm 0.9\%$ for a reflectivity of $48.4 \pm 0.5\%$, and Fig. 6*c* showing a visibility of $12.9 \pm 0.9\%$ for a reflectivity of $94.1 \pm 0.2\%$. The fit to Fig. 6*a* is the calculated visibility for the given coupling ratio. This device has also been used to investigate one-, two- and four-photon interference pattern for applications in quantum metrology [30].

This simple variable beamsplitter element can be used as a building block for constructing arbitrarily large $N \times N$ quantum optical circuits capable of implementing any unitary operation on many waveguide modes. A similar scheme was first proposed by Reck *et al.* [32], demonstrating that a general multimode interferometer could be constructed using a triangular array of beamsplitters and phase shifters. A reconfigurable bulk optic circuit implementation of this form could only be realised for a small number of modes, due to issues of stability and circuit complexity. But, significantly larger circuits could be possible in waveguide circuits by replacing each variable beamsplitter element with the MZI in Fig. 5. Fig. 7 shows a schematic diagram of how this type of circuit could be implemented in waveguides for a 7×7 arbitrary unitary operation. Examples of large-scale

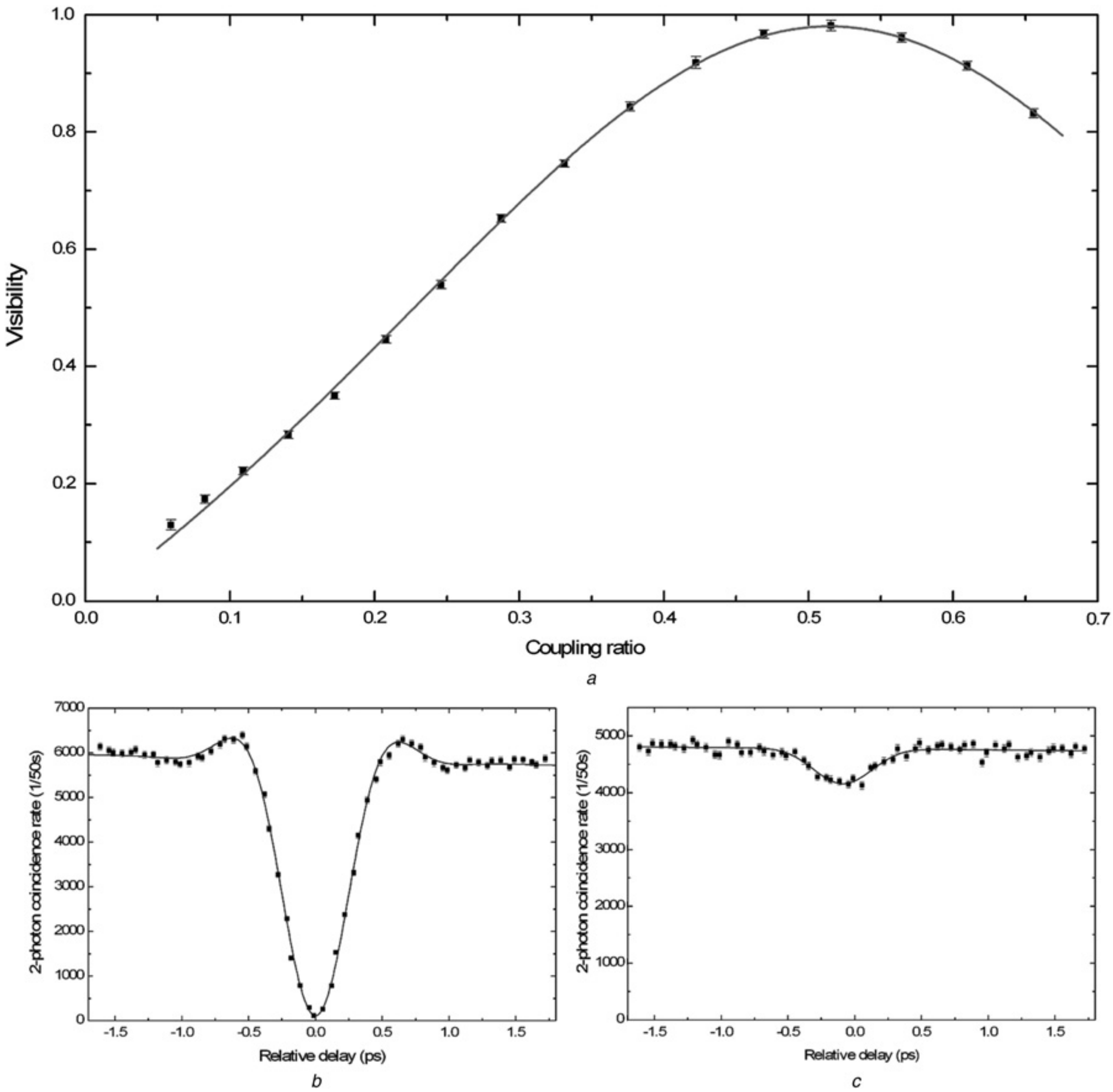

Fig. 6 *Non-classical interference for a variable beamsplitter*

a Visibility of two-photon coincidence dip plotted against coupler reflectivity of a variable beamsplitter
b Two-photon coincidence plot for 48.4% reflectivity and
c Two-photon coincidence plot for 94.1% reflectivity

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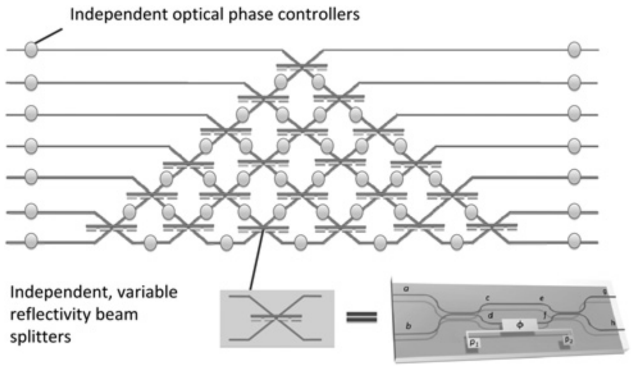

Fig. 7 *Construction of an $N \times N$ arbitrary unitary from phase shifters and variable beamsplitters*

thermo-optical switches have been demonstrated for use in telecommunications applications, with a 32×32 waveguide switch comprising of 2048 individual elements [33]. Implementing an arbitrary unitary on this number of modes would require a similar number of reconfigurable 2×2 elements, and a device of this scale would be inconceivable in bulk optics.

The thermo-optical switch allows reconfiguration times on the millisecond scales, and although the phase can be controlled with good accuracy, fast switching times cannot be obtained through changes in temperature. For this reason thermo-optical manipulation of integrated phases can only be used for quantum processes that do not require ultra-fast operation, including state preparation, quantum measurement and perhaps even full-scale quantum computing [34]. Other applications demand fast switching, such as adaptive circuits for quantum control and feed-forward processes such as the ones required in cluster state computation [35, 36]. These applications will require sub-nanosecond switching, possible with electro-optic materials such as lithium niobate ($LiNbO_3$) (used to make high speed modulators for telecommunications applications operating at tens of gigahertz [37]).

7 Two-photon quantum logic

A fully universal quantum computer can be constructed using only arbitrary single-qubit rotations (similar to the devices presented in the previous sections) and any non-trivial two-qubit operation, such as the controlled-NOT (CNOT) gate. So in principle, a photonic quantum computer could be fabricated by integrating on the same circuit reconfigurable couplers, phase shifters and CNOT devices – assuming that suitable sources and detector were available. A two-qubit operation in quantum optics allows the state of one photon to affect the state of another photon, and in the case of the CNOT gate, it specifically allows the states of two photons to become entangled. In effect it is an entangling gate. The CNOT gate has two inputs, the control (C) qubit, which passes unchanged through the device, and the target (T) qubit, which flips state conditional on the control (C) qubit being in the $|1\rangle$ state. The schematic representation and truth table of the CNOT are presented in Fig. 8.

In principle, the CNOT gate can be implemented using optical devices with a highly non-linear element to support the required photon–photon interaction, but to date a non-linear interaction of the required strength has not been realised [14]. As an alternative approach, using the properties of non-classical interference on a 1/3 beamsplitter, the required phase shift can be induced with a probability of occurrence of 1/9, and a non-deterministic CNOT gate can be realised using only linear optical components [38, 39]. The waveguide circuit layout of the CNOT gate is presented in Fig. 8, and is essentially a direct implementation of the theoretical scheme presented in [39], comprising two 1/2 couplers and three 1/3 couplers. The control and target qubits are each encoded by a photon in two waveguides, and the success of the gate is heralded by detection of a photon in both the control and target outputs, which happens with probability of 1/9.

The gate works as follows: control C and target T qubits are each encoded by photons in the dual-rail representation. The two 1/2 splitters in the target waveguides (T_0 and T_1) form a balanced MZI, so that, with no inputs in the control waveguides (i.e. a photon in the control state zero C_0), the target photons exit the interferometer from the same port as the input port. For the control in the state one (photon in the waveguide C_1), the control and target photons interfere non-classically at the central 1/3 splitter. For such a value of splitting ratio, the result state evolves to $|1\rangle_C|0\rangle_T \rightarrow -1/3(|1\rangle_C|0\rangle_T)$. The negative sign indicates a π-phase difference between the arms of the interferometer, thus causing the target qubit to be flipped, exiting the opposite port of the interferometer. The presented scheme does not always work; it is possible, for example, to find two photons in the control or target outputs. However, the presence of only one photon in the control and one photon in the target herald the success of the gate. The probability of such an event is equal to 1/9 for all the possible combinations of inputs states.

The truth table for the operation of this device in the computational base states is presented in Fig. 9, comparing the ideal case (left plot) and the actual measured results (right plot). For the conditions with the control qubit set to $|0\rangle_C$, there is an excellent agreement with the ideal case, with peak values of 98.5%. This is a measure of the

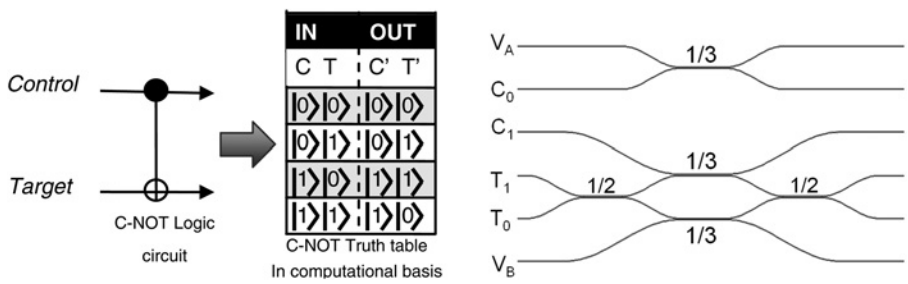

Fig. 8 *Controlled-NOT gate*

Left: schematic representation of the CNOT gate and corresponding truth table; right: waveguide circuit implementation of CNOT gate

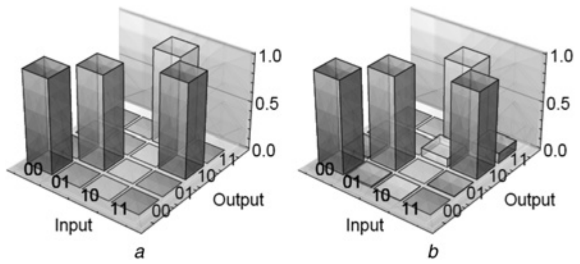

Fig. 9 *Characterisation of integrated CNOT gate for*

a Ideal truth table
b Measured truth tables in the computational basis

classical interference within the interferometer of the device and demonstrates the excellent stability and sub-wavelength scale length control on the fabrication of these waveguide devices. The average of all the logical basis fidelities for this device is $F = 94.3 \pm 0.2\%$, showing high fidelity operation of the CNOT gate in the computational basis. To obtain a complete characterisation of the quantum gate the fidelity value is not sufficient, and a full process tomographic study is required. This requires many additional MZIs and phase shifters to implement, and at the time of publication such a device had not been characterised.

8 Shor's quantum factoring algorithm

In the previous sections, the key building blocks for single- and two-qubit operations have been introduced. The next stage is to use these building blocks to form more complicated quantum circuits. One such example that has been recently demonstrated is the on-chip implementation of the compiled Shor's factoring algorithm [40].

As one of the main motivational drives for developing quantum technologies, it is anticipated that computation via quantum algorithms will significantly outperform classical computers. One such algorithm is Shor's factoring algorithm, which was proposed 15 years ago by Peter Shor [5]. The algorithm harnesses the unique quantum mechanical properties of superposition and entanglement to efficiently factorise a product of two large prime numbers. To date, no known classical algorithm exists to solve this problem in time less than exponentially large in the input number, making factorisation intractable and forming the basis for most modern cryptographic security systems. Originally conceived in the context of simulating complex quantum systems, Shor's quantum factoring algorithm showed the possibility of factoring the product of two large prime numbers exponentially faster than any known conventional method.

Progress towards proof-of-principle demonstration of Shor's factoring algorithm have included liquid-state nuclear magnetic resonance [41] and bulk optical implementations using simplified quantum logic gates [42, 43]. While the latter demonstrated the necessary use of entanglement required for improvement over classical computation, they cannot so far be scaled to the required large number of quantum gates and qubits needed to factor usefully large numbers. Instability and physical size remain among the largest of the limitations in complexity in implementing practical quantum technologies. Here, a compiled version of Shor's factorise 15 is described. The circuit operates on four qubits in which the processing occurs in a photonic circuit of several one- and two-qubit gates similar to those presented in the previous sections. Whereas the full Shor's algorithm is designed to factorise any given input, the compiled version is designed to find the prime factors of a specific input.

The full process of Shor's algorithm can be partitioned into conventional classical processing and a quantum sub-routine known as order finding [5, 42, 43]. Fig. 10*a* shows the quantum circuit for the compiled order finding routine designed specifically to factorise 15, with a designed success probability of 0.5. The circuit comprises a number of single qubit rotations (Hadamard, NOT [X]) and two two-qubit entangling gates (controlled π-phase shift [C-Z]) acting on qubits in two registers known as the argument (xi) and function (fi). An inverse quantum Fourier transform (QFT) acting on the argument register is achieved for certain compiled circuits using classical post-processing [42]. The integrated waveguide version of this circuit is shown in Fig. 10*b*, and consists of six Hadamard gates and two controlled π-phase gates. A picture of the actual chip used to perform the measurements is shown in Fig. 11.

The circuit requires four photonic quantum bits, encoded using the path of four 790 nm photons simultaneously prepared via spontaneous parametric down-conversion (SPDC). The photons are injected into the waveguide chip using buttcoupled arrays of polarisation maintaining fibre, and detected after the waveguide circuit using single-photon counting avalanche photodiodes (APDs) to yield the outcome of the quantum algorithm. The state $|\psi_{in}\rangle$ is used at the input and the output state of qubits x_1 and x_2 are measured. The output statistics shown in Fig. 10*c* give the four binary outcomes: 000, 010, 100 and 110 (including the x_0 qubit). Outputs 010 and 110 lead to the correct calculation for finding the order $r = 4$ for the algorithm, which then enables efficient classical computation of the factors 3 and 5; 100 gives the trivial factors (1 and 15); and

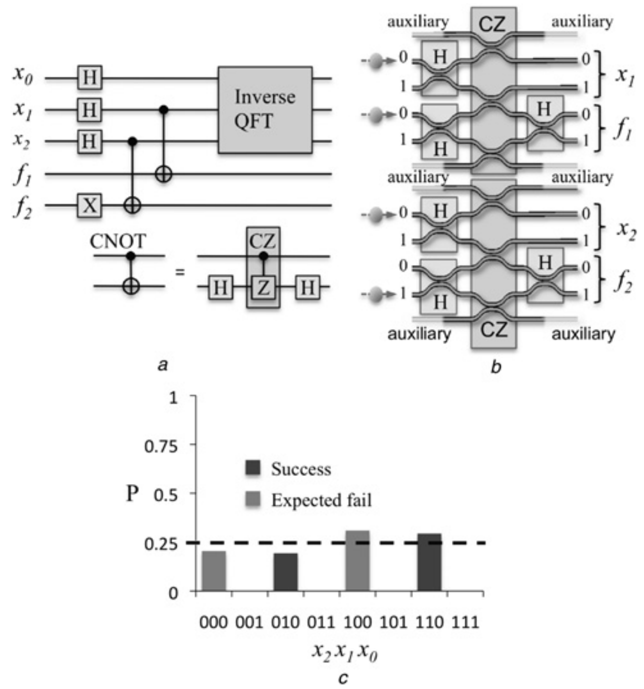

Fig. 10 *Integrated optical implementation of Shor's quantum factoring algorithm*

a Quantum circuit, comprising Hadamard (H), CNOT gates the QFT
b Schematic of the quantum waveguide circuit
c Results from the algorithm. The dashed line corresponds to the ideal probability distribution

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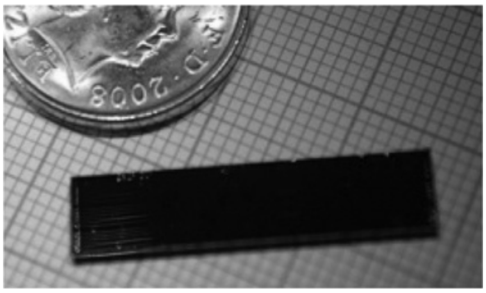

Fig. 11 *Integrated quantum photonic circuits used to implement the compiled version of Shor's quantum factoring algorithm*

000 is an expected failure mode inherent to Shor's algorithm. The measured results have a fidelity of $99 \pm 1\%$ with the ideal probability distribution. This demonstration of a small-scale compiled Shor's algorithm on a chip shows promise for quantum computing in integrated waveguides.

9 Outlook

Achieving large-scale and reliable computation based on quantum physics would have a significant impact on the way information is processed, encoded and transmitted, spawning an anticipated second information revolution [8]. The work highlighted in this paper and the current advancements in integrated quantum optics draw parallels with the very initial developments in the field of integrated electronics in the 1960s, when electronic circuits moved from discrete transistors, resistors and capacitors on an electronic breadboard, towards the sophisticated, powerful and fully integrated computer processing units that are readily available today. The field of quantum optics is still in its very early days, and the research on integrated quantum photonics marks the first step in moving from discrete optical components on an optical table, to integrated optical circuits capable of performing simple quantum information processing operations, paving the way for the development of future advanced quantum information processing technologies.

To date, all demonstrations of integrated quantum photonic circuits have relied on external non-scalable single-photon sources and modest efficiency single-photon detectors. However, to achieve full integration, and thereby realise useful circuits that can exist outside of the research laboratory, an integration technology is required that can include single-photon sources, detectors and waveguides on a single optical chip. A strong optical non-linearity at the single photon level is also desirable to mediate on-chip light-matter interactions of photons.

Indistinguishable pairs of photons generated through SPDC (similar to the setup shown in Fig. 4) have been an invaluable tool for proof-of-principle experiments in quantum optics. Periodically poled lithium niobate [44] and silicon nanowires [45] are two examples where photon pairs have been successfully generated within waveguide structures. Unfortunately, these types of sources do not scale well, and are unsuitable for full-scale quantum computing. The ideal photon source should have high efficiency, a very small probability of emitting more than one photon per pulse, produce indistinguishable photons on demand and couple efficiently to waveguides [46]. Promising alternatives to SPDC are addressing these stringent requirements, including sources based on quantum dots [47], colour-centres in diamond [48] and impurities in a semiconductor [46]. An even greater challenge is achieving strong non-linear interactions on-chip between two photons, a process which is needed for deterministic (non-probabilistic) two-photon logic gates. Great progress has been made using photonic crystal nanocavities containing a strongly coupled quantum-dots, in which photon-induced phase shifts of up to $\pi/4$ and amplitude modulation of up to 50% have been achieved [49].

The development of high-efficiency detectors is an important step for the advancement of quantum optic technologies [50]. Currently, commercially available single-photon detectors based on APDs are the typical choice for quantum optic experiments since they have reasonable efficiencies of $\sim$70% (for wavelengths around 800 nm), but even higher efficiencies will be required for scalable optical quantum computing. Detectors based on superconducting nanowires are showing significant progresses [51], and promise a route towards higher efficiency in the longer wavelength range, fast response speed and photon number resolving capabilities. Already, the first application of superconducting detectors to integrated quantum photonic circuits was demonstrated [52], allowing the measurement of quantum interference effects with better visibility than when using standard silicon APDs.

In addition to the development of highly efficient single-photon sources, detectors and optical non-linearities for integration with waveguide circuits, the waveguide circuits themselves must be further developed. Circuits with additional capabilities, like ultra-fast electro-optic control and polarisation diversity will expand the range of possible applications by allowing fast switching and reconfigurable circuits. Further miniaturisation will be possible with higher index contrast waveguides, by using material systems such as InP, GaAs and silicon, and by employing nanowire and photonics crystal waveguide structures.

10 Conclusions

This paper has reviewed a new approach in the way quantum optic information science experiments are conducted. The key building blocks of one- and two-qubit gates have been presented, and routes to realising large-scale integrated reconfigurable circuits have been discussed. These components form the building blocks required to realise more complex quantum integrated circuits, and by combining the operation of multiple gates on a single chip, a compiled version of Shor's factoring algorithm was realised, using four qubits, six Hadamard gates and two controlled phase gates.

For quantum optic technologies to flourish, an integration approach must be taken and further developed to include integration of efficient photon sources, photon detector and photon–photon non-linear interactions to enable the development of compact and efficient quantum optic components with unprecedented complexity. This on-chip control of light at the quantum level will enable new scientific developments in the field of quantum optics, as well as the realisation of advanced quantum systems for the purposes of quantum metrology, quantum communications and quantum computation.

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12 References

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